# Biomedical Named Entity Recognition with BioB

Lead Author: T. Harshini Priya (Lead)

Faculty Guide: Dr. K. V. Ramanathan, M.Tech, Ph.D.

Institutional Department: Department of Data Science

## 1. PROJECT OVERVIEW & SCOPE

Fine-tuned BioBERT transformer model with conditional random fields [CRF] for automated extraction of genes, diseases, and chemical compounds from scientific literature.

## 2. EXTRACTION METRICS

Achieved 89.7% F1-score across NCBI-Disease and BC5CDR gold standard corpora with sub-20ms inference latency.

## 3. REAL-TIME API

Integrated with FastAPI REST endpoints for real-time indexing of PubMed biomedical abstracts.